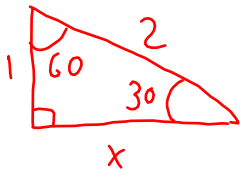
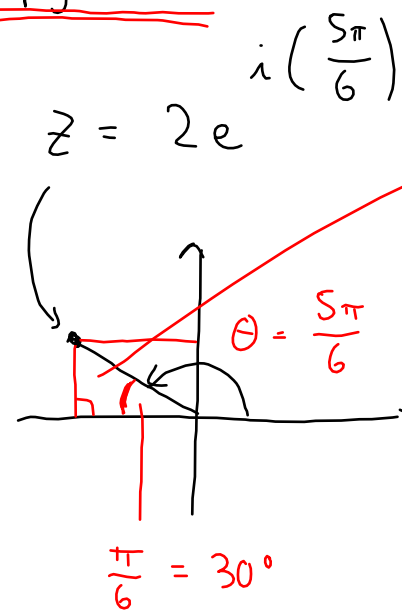


Løsningsforslag midtreis prøveeksamen MAT 1100
4. oktober 2025

Oppgave 1



$$\begin{aligned}1^2 + x^2 &= 2^2 \\x^2 &= 3 \\x &= \sqrt{3}\end{aligned}$$

$$z = -\sqrt{3} + i$$

E

Oppgave 2

$$\begin{aligned}z^2 + 2z + 2 &= 0 \quad \text{gir} \quad z = \frac{-2 \pm \sqrt{4 - 8}}{2} = \frac{-2 \pm \sqrt{-4}}{2} \\&= \frac{-2 \pm \sqrt{-1} \cdot 2}{2} = \frac{-2 \pm \sqrt{-1} \cdot \sqrt{4}}{2} \\&= \frac{-2 \pm i \cdot 2}{2} = -1 \pm i\end{aligned}$$

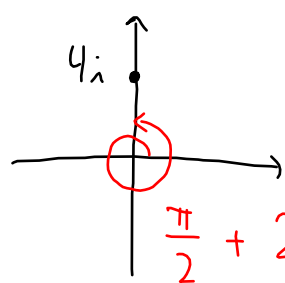
D

Oppgave 3

E $w = i$ er en femterot av $z = i$, fordi

$$w^5 = i^5 = (i^2)^2 \cdot i = (-1)^2 \cdot i = 1 \cdot i = i$$

Oppgave 4



$$4 = re^{i\theta} = 4e^{i\left(\frac{5\pi}{2}\right)}$$

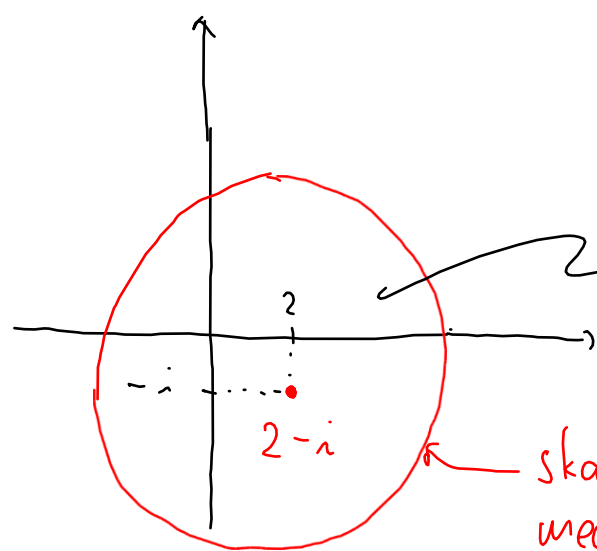
$$\frac{\pi}{2} + 2\pi = \frac{\pi}{2} + \frac{4\pi}{2} = \frac{5\pi}{2}$$

B

Oppgave 5

$$|2 - z - i| < 5 \text{ kan skrives } |(2-i) - z| < 5$$

dvs. kravet er at avstanden mellom punktet $2-i$ og punktet z skal være mindre enn 5.



z må ligge innenfor denne sirkelen

skal forestille en sirkel med sentrum $2-i$ og radius 5

C

Oppgave 6

$$\sum_{n=1}^2 \left(\sum_{j=1}^{100} (\sqrt{j} - \sqrt{j-1}) \right)^n$$

$$= \sum_{n=1}^2 \left((\cancel{\sqrt{1}} - \underbrace{\sqrt{0}}_0) + (\cancel{\sqrt{2}} - \cancel{\sqrt{1}}) + (\cancel{\sqrt{3}} - \cancel{\sqrt{2}}) + \dots + (\sqrt{100} - \cancel{\sqrt{99}}) \right)^n$$

$$= \sum_{n=1}^2 (\sqrt{100})^n = \sum_{n=1}^2 10^n = 10^1 + 10^2 = 10 + 100 = 110$$

A

Oppgave 7

Deler på dominerende ledd

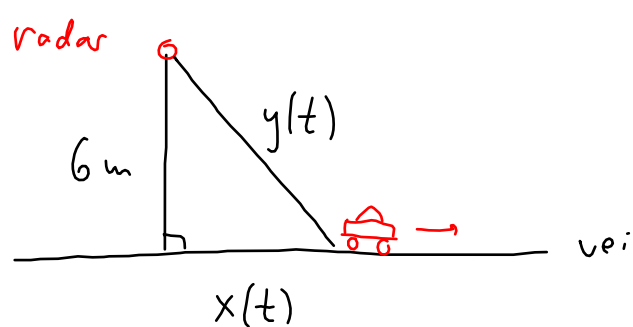
$$\lim_{n \rightarrow \infty} \frac{(-n)^5 - (-6)^n + \pi^n}{1 + (-10)^n} = \lim_{n \rightarrow \infty} \frac{\frac{(-n)^5}{(-10)^n} - \frac{(-6)^n}{(-10)^n} + \frac{\pi^n}{(-10)^n}}{\frac{1}{(-10)^n} + 1}$$

$$= \lim_{n \rightarrow \infty} \frac{-\frac{n^5}{(-10)^n} - \left(\frac{-6}{-10}\right)^n + \left(\frac{\pi}{-10}\right)^n}{\left(-\frac{1}{10}\right)^n + 1} = \frac{0}{1} = 0$$

$\left(\frac{6}{10}\right)^n \rightarrow 0$

C

Oppgave 8



$$(x(t))^2 + 6^2 = (y(t))^2 \quad (\text{Pyt.})$$

Implisitt derivasjon:

$$\cancel{2} \cdot x(t) \cdot x'(t) = \cancel{2} \cdot y(t) \cdot y'(t)$$

Tidspunktet vårt: $x = 8$, så $y^2 = 6^2 + 8^2 = 36 + 64 = 100$
dvs. $y = 10$. Dessuten $y'(t) = 80 \text{ km/h}$
 $= \frac{80}{3,6} \text{ m/s}$

Innsatt:

$$8 \cdot x'(t) = 10 \cdot \frac{80}{3,6}$$

$$x'(t) = \frac{800}{8 \cdot 3,6} = \frac{100}{3,6} \text{ m/s} = \underline{100 \text{ km/h}}$$

E

Oppgave 9

$$a_0 = 0 \quad \text{og} \quad a_{n+1} = \frac{3a_n - 12}{9} \quad \text{for } n \geq 0$$

Hvis følgen konvergerer mot L , må vi ha $L = \frac{3L - 12}{9}$

Dette gir $9L = 3L - 12$, $6L = -12$, $L = -2$

Anta at $a_n > -2$. Da får vi

$$a_{n+1} = \frac{3a_n - 12}{9} > \frac{3 \cdot (-2) - 12}{9} = \frac{-6 - 12}{9} = -2$$

Da har vi vist ved induksjon at $a_n > -2$ for alle n . Dette gir

$$a_{n+1} = \frac{3a_n - 12}{9} = \frac{3a_n + 6 \cdot (-2)}{9} < \frac{3a_n + 6a_n}{9} = \frac{9a_n}{9} = a_n$$

Så følgen er avtakende og nedad begrenset av -2 . Altså konvergerer den, ved komplotthetsegenskapen for følger. Da må grensen være -2 . D

Oppgave 10

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{x \ln x}{3^x} &\stackrel{[\frac{\infty}{\infty}]}{=} \lim_{x \rightarrow \infty} \frac{x \cdot \ln x}{e^{(\ln 3)x}} \stackrel{[\frac{\infty}{\infty}]}{=} \lim_{x \rightarrow \infty} \frac{1 \cdot \ln x + x \cdot \frac{1}{x}}{e^{(\ln 3)x} \cdot (\ln 3)} \\ &= \lim_{x \rightarrow \infty} \frac{\ln x + 1}{e^{(\ln 3)x} \cdot (\ln 3)} \stackrel{[\frac{\infty}{\infty}]}{=} \lim_{x \rightarrow \infty} \frac{\frac{1}{x} \rightarrow 0}{\underbrace{e^{(\ln 3)x}}_{\rightarrow \infty} \cdot (\ln 3)^2} = 0 \end{aligned}$$

$3^x = (e^{\ln 3})^x = e^{(\ln 3)x}$

C

Oppgave 11

$$\begin{aligned} f(x) &= \int_1^{ax} (t^2 - 2t) dt = \left[\frac{1}{3} t^3 - t^2 \right]_{t=1}^{t=ax} \\ &= \left[\frac{1}{3} (ax)^3 - (ax)^2 \right] - \left[\frac{1}{3} \cdot 1 - 1 \right] = \frac{1}{3} a^3 x^3 - a^2 x^2 + \frac{2}{3} \end{aligned}$$

$$\text{der. } f(2) = \frac{8}{3} a^3 - 4a^2 + \frac{2}{3} \quad \boxed{E}$$

Oppgave 12

$$\lim_{x \rightarrow 0^+} (2 \sin x)^{3x} \stackrel{[0^0]}{=} \lim_{x \rightarrow 0^+} \left(e^{\ln(2 \sin x)} \right)^{3x} = \lim_{x \rightarrow 0^+} e^{3x \cdot \ln(2 \sin x)}$$

EkspONENTEN:

$$\begin{aligned} \lim_{x \rightarrow 0^+} 3x \cdot \ln(2 \sin x) &\stackrel{[0 \cdot \infty]}{=} \lim_{x \rightarrow 0^+} \frac{3 \ln(2 \sin x)}{\frac{1}{x}} \stackrel{[\frac{\infty}{\infty}]}{=} \lim_{x \rightarrow 0^+} \frac{3 \cdot \frac{1}{2 \sin x} \cdot \cancel{2} \cos x}{-\frac{1}{x^2}} \\ &= -3 \lim_{x \rightarrow 0^+} \frac{x^2 \cos x}{\sin x} \\ &\stackrel{[\frac{0}{0}]}{=} -3 \lim_{x \rightarrow 0^+} \frac{2x \cos x - x^2 \sin x}{\cos x} = -3 \cdot \frac{0}{1} = 0 \end{aligned}$$

$$\text{Så } \lim_{x \rightarrow 0^+} (2 \sin x)^{3x} = e^0 = 1. \quad \boxed{C}$$

Oppgave 13

$$f: (0,1) \rightarrow \mathbb{R} \text{ ved } f(x) = \ln((\ln x)^2)$$

$$f'(x) = \frac{1}{(\ln x)^2} \cdot 2 \ln x \cdot \frac{1}{x} = \frac{2}{x \cdot \ln x}$$

$$f''(x) = \frac{0 - 2(1 \cdot \ln x + x \cdot \frac{1}{x})}{(x \cdot \ln x)^2} = \frac{-2(\ln x + 1)}{(x \ln x)^2}$$

Fortegnet til $f''(x)$ bestemmes av faktoren $(\ln x + 1)$

$$\ln x + 1 = 0 \text{ gir } \ln x = -1, e^{\ln x} = e^{-1}, x = e^{-1} = \frac{1}{e}.$$

A

Oppgave 14

$$f: \mathbb{R} \rightarrow \mathbb{R} \text{ ved } f(x) = \arctan x - ax$$

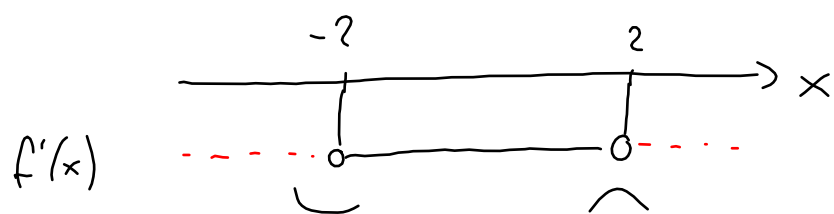
$$f'(x) = \frac{1}{1+x^2} - a = \frac{1}{1+x^2} - \frac{a(1+x^2)}{1+x^2} = \frac{1 - (a + ax^2)}{1+x^2}$$

$$= \frac{1 - a - ax^2}{1+x^2}$$

$$f'(2) = 0 \text{ gir } 1 - a - 4a = 0$$

$$1 - 5a = 0, a = \frac{1}{5}$$

$$\text{Med } a = \frac{1}{5} \text{ har vi } f'(x) = \frac{\frac{4}{5} - \frac{1}{5}x^2}{1+x^2} = \frac{1}{5} \frac{4 - x^2}{1+x^2}$$



Nå har f et lokalt
maksimalpunkt i $x = 2$

E

Oppgave 15

$$(f^{-1})'(f(x)) = \frac{1}{f'(x)} \quad x = 2$$

$$(f^{-1})'(f(2)) = \frac{1}{f'(2)}$$

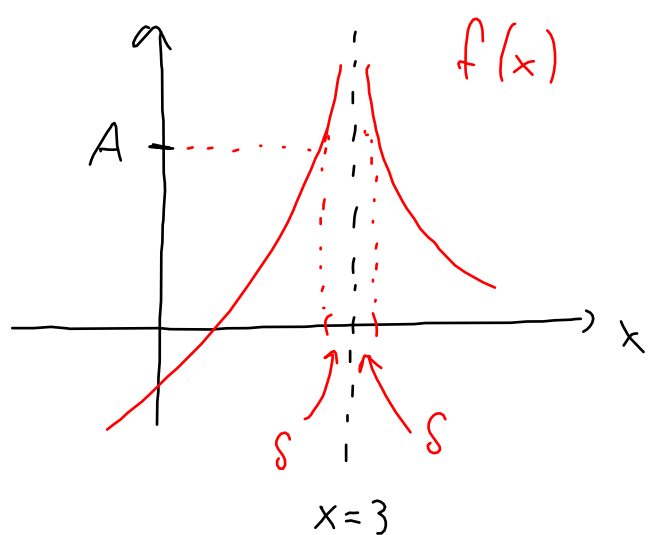
Siden $f^{-1}(7) = 2$, må $f(2) = 7$. Dette gir

$$(f^{-1})'(7) = \frac{1}{f'(2)} \quad \text{Altså} \quad \frac{1}{3} = \frac{1}{f'(2)}, \text{ så } f'(2) = 3$$

C

Oppgave 16

$f: \mathbb{R} \rightarrow \mathbb{R}$, definisjon av $\lim_{x \rightarrow 3} f(x) = +\infty$?



D Riktig.

Oppgave 17

$$f(x) = x \cdot 10^{-\frac{5}{x}} = \frac{x}{e^{(\ln 10) \cdot \frac{5}{x}}}$$

hvis du vil

Tester for skråasymptote:

$$a = \lim_{x \rightarrow \infty} \frac{f(x)}{x} = \lim_{x \rightarrow \infty} \frac{1}{e^{(\ln 10) \cdot \frac{5}{x}}} = \frac{1}{e^0} = 1$$

$$b = \lim_{x \rightarrow \infty} [f(x) - ax] = \lim_{x \rightarrow \infty} (x \cdot 10^{-\frac{5}{x}} - x)$$

$$= \lim_{x \rightarrow \infty} x \cdot \left(e^{(\ln 10) \cdot (-\frac{5}{x})} - 1 \right)$$

$$= \lim_{x \rightarrow \infty} \frac{e^{-(\ln 10) \cdot \frac{5}{x}} - 1}{\frac{1}{x}} = \lim_{x \rightarrow \infty} \frac{e^{-\cancel{(\ln 10) \cdot 5 \cdot \frac{1}{x}}}}{-\frac{1}{x^2}} \quad \left[\frac{0}{0} \right]$$

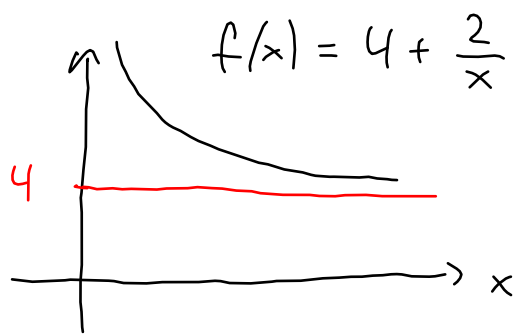
$$= \frac{e^0 \cdot (-5 \ln 10)}{1} = -5 \ln 10$$

Skråasymptote $y = ax + b = \underline{x - 5 \ln 10}$

D

Oppgave 18

$$f(x) = 4 + \frac{2}{x}$$



$$|f(x) - 4| = \left| \left(4 + \frac{2}{x} \right) - 4 \right| = \left| \frac{2}{x} \right| = \frac{2}{x} < \varepsilon$$

hvis $2 < \varepsilon x$

da $\frac{2}{\varepsilon} < x$.

B